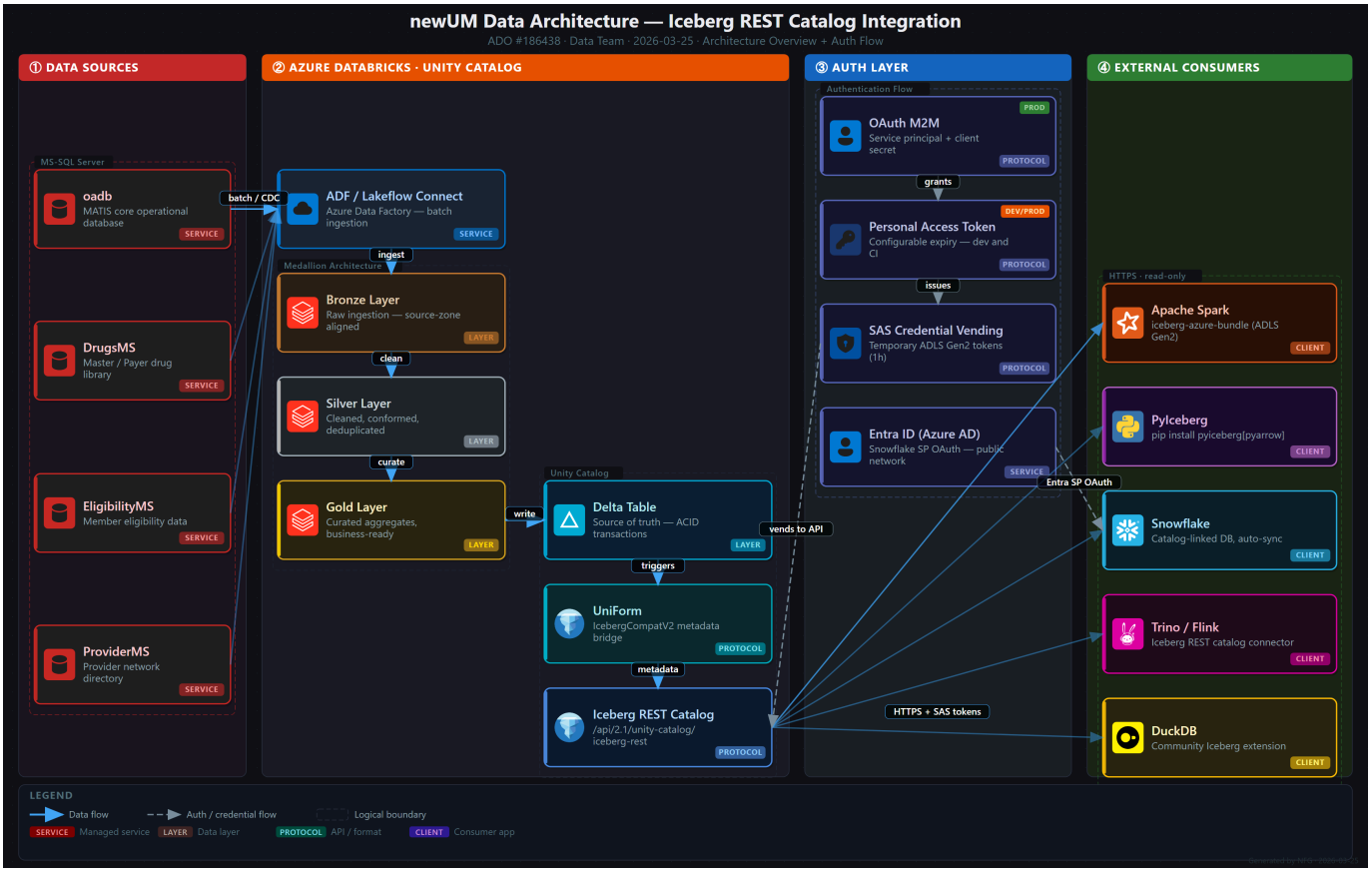
Closure Note — Investigate Iceberg REST Catalog API Feasibility

ADO Ticket: #186438 **Area**: newUM\Data Team **Sprint**: TBD **Priority**: P2 **Sources**: All claims verified against official Microsoft Learn docs (fetched 2026-03-23) AND validated against TEST workspace API (2026-03-24). See References for full list. **Project context**: `../../knowledge.json` v1.14.0 — operational facts cited as [K].

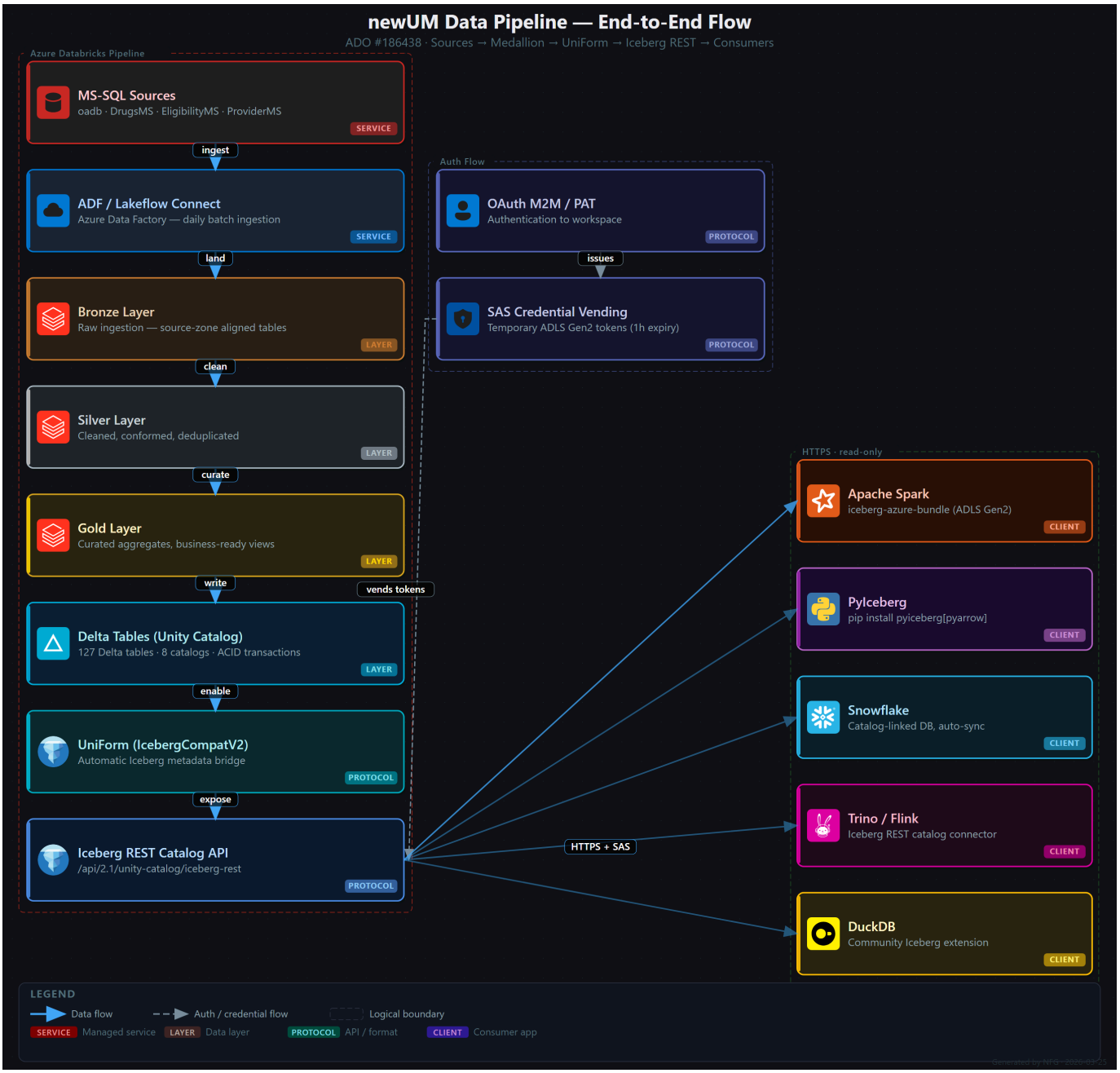
Architecture Overview



Data flow: Writes go through Databricks (Delta) → UniForm generates Iceberg metadata async → External clients read via REST Catalog API → Credential vending provides temporary ADLS SAS tokens.

> *Sources: [S1] endpoint + credential vending; [S2] UniForm async metadata generation.*

Applied to newUM



The above architecture maps to our confirmed environment [K: `tech_stack.data`, `environments.databricks`]:

Layer	newUM Reality	Source
Delta tables	Unity Catalog with Bronze/Silver/Gold medallion	[K: `tech_stack.data`]
Workspace (DEV)	`https://adb-3806388400498653.13.azuredatabricks.net/` — BLOCKED (contains PHI)	[K: `databricks.workspaces[0]`]
Workspace (TEST)	`https://adb-2393860672770324.4.azuredatabricks.net/` — access GRANTED via ticket #0035611 (2026-03-23)	[K: `databricks.workspaces[1]`]
Workspace (UAT/PROD)	URLs unknown	[K: `databricks.workspaces[2-3]`]
PAT token	`visualstudio-carlos` created, expires 2027-03-22	[K: `access_inventory.results["Databricks test"]`]
Auth method	Entra ID button click required (Okta SSO does NOT auto-login to Databricks)	[K: `access_status.access_inventory.note`]
Data Team Lead	Michal Mucha — runs daily standups, Databricks Lakeflow Connect chat	[K: `key_contacts["Michal Mucha"]`]
DevOps contact	`devopsrequest@oncologyanalytics.com`	[K: `onboarding_documents.devops_email`]

Target databases	oadb (main), DrugsMS, EligibilityMS, ProviderMS — all MS-SQL, candidates for lakehouse migration	[K: `environments.databases`]

Cost Analysis — Is It Really Zero?

Short answer: **The Iceberg REST Catalog API itself has zero incremental licensing cost.** But that deserves a breakdown, because "zero" is surprising and the boss is right to question it.

Why It Appears Free

The Iceberg REST Catalog API is **not a separate service** — it's a built-in endpoint within the existing Databricks workspace. Think of it like enabling HTTPS on a database that's already running. There is no:

- New compute cluster to provision
- New storage account to create
- New license tier to purchase
- Per-API-call billing (unlike, say, Azure API Management or AWS API Gateway)

This is a Databricks product strategy decision: they include it to make Unity Catalog more attractive vs. competitors (Snowflake, BigQuery) by positioning Databricks as the universal data lakehouse. The API is the "front door" — they want you using it.

What Actually Costs Money (the fine print)

Component	Cost	Who Pays	Already Paying?
Databricks workspace	DBU pricing (existing)	OncoHealth	**YES** — already running
Delta writes (trigger metadata gen)	Same cluster as current writes	OncoHealth	**YES** — writes happen regardless
Iceberg metadata files	~KB per table version on ADLS	OncoHealth	**YES** — marginal, within existing storage
Driver memory overhead	+5-15% on write operations ([S2])	OncoHealth	**YES** — uses existing cluster
Service principal (OAuth M2M)	$0 — Entra/Databricks feature	—	N/A
SAS credential vending	$0 — built into API response	—	N/A
Network egress (ADLS → reader)	Azure egress pricing if cross-region	Depends on reader location	**Only if cross-region**
Private Link (if required)	~$7.30/month + $0.01/GB	OncoHealth	**Only if private networking needed**

The Real Cost: Operational, Not Financial

- **One-time setup**: Enable UniForm on target tables, grant EXTERNAL_USE_SCHEMA, enable external_access on metastore → ~2-4 hours of data engineer time
- **Ongoing**: Monitor metadata lag, ensure write clusters stay on DBR 14.3+ LTS → negligible if already on modern runtime
- **Risk**: Protocol upgrade is partially irreversible (Delta reader/writer version bumps can't be undone)

Why Not Just Read Delta Lake Directly?

This is the key architectural question. Delta Lake already has ACID transactions, schema evolution, and time travel — so it's not the same as reading raw Parquet. The issue is what happens when systems **outside Databricks** need to read that data.

Dimension	Direct Delta Lake Read	Iceberg REST Catalog API
Client compatibility	Requires Delta-compatible library (Spark, delta-rs, Delta Sharing). Our .NET backend doesn't speak Delta natively.	Open standard — PyIceberg, Spark, Snowflake, Trino, DuckDB all have native Iceberg REST connectors
Auth	Need ADLS storage keys or SAS tokens — must generate and rotate manually	API-level auth (PAT/OAuth) → automatic SAS credential vending with 1h expiry
Governance	Direct storage access bypasses Unity Catalog — no audit trail, no column masking, no row filters	All reads go through UC permissions → audit logs, column-level security, row filters

Schema	Must parse `_delta_log/` JSON to understand schema, partitions, current files	Standard catalog protocol — schema discovery built-in
ACID isolation	Risk of reading mid-write ("dirty reads") if not using a Delta-aware client	Iceberg metadata guarantees snapshot isolation — always reads a consistent version
Deletion Vectors	Must understand Delta's DV format to skip deleted rows — most non-Spark readers can't	Handled transparently by Iceberg metadata layer
Column Mapping	Delta's column ID mapping means physical Parquet column names ≠ logical names — non-Delta readers return wrong column names	Iceberg metadata maps logical ↔ physical correctly
Time Travel	Must manually parse Delta log for historical versions	Iceberg snapshots supported natively
Security Compliance	Gives external systems direct ADLS access — storage-level blast radius	API-level access only — temporary SAS tokens (1h), scoped to specific tables

Bottom line: Delta Lake is the internal language of our lakehouse. Iceberg REST is the translator that lets any external system read it — without needing a Delta-aware library, without us handing over our storage keys, and with full audit trail. For a healthcare company handling PHI-adjacent data, the governance difference alone justifies the approach.

> *Sources: [S1] credential vending + audit; [S2] UniForm metadata generation + resource impact; [S2] "might increase the driver resource usage".*

Investigation Files

- Full report: `output.md`

Pros

- **Zero data duplication** — Iceberg reads use same Parquet files as Delta; only metadata is generated ([S2]: "*A single copy of the data files serves multiple formats*")
- **Native Azure Databricks support** — endpoint is built-in, no external infrastructure needed ([S1]: "*Unity Catalog provides an implementation of the Iceberg REST catalog API*")
- **Broad client compatibility** — PyIceberg, Spark, Snowflake, Trino, Flink supported per [S1]; DuckDB via its own Iceberg extension (not in Databricks docs)
- **Credential vending** — temporary SAS tokens issued automatically, default 1h expiry ([S1]: "*The default expiration time is one hour*")
- **Low operational overhead** — metadata generation is automatic and async ([S2]: "*asynchronously after a Delta Lake write transaction completes*")
- **Standards-based** — uses official Apache Iceberg REST Catalog spec ([S1])
- **Public Preview (DBR 16.4+)** — on GA track, production-viable with preview caveats ([S1]: "*Public Preview in Databricks Runtime 16.4 LTS and above*")

Cons

- **Read-only for Delta+UniForm tables** — external clients cannot write; writes must go through Databricks ([S2]: "*Iceberg client support is read-only. Writes are not supported.*")
- **Metadata lag** — Iceberg metadata generated asynchronously; may lag behind latest Delta version ([S2]: "*Delta table versions do not align with Iceberg versions*")
- **Protocol upgrade partially irreversible** — Iceberg reads can be disabled by unsetting `delta.universalFormat.enabledFormats`, but Delta protocol version upgrades and column mapping **cannot** be undone ([S2]: "*You can turn off Iceberg reads by unsetting the delta.universalFormat.enabledFormats table property. Upgrades to Delta Lake reader and writer protocol versions cannot be undone.*")
- **Deletion vectors incompatible with Iceberg v2** — tables need `REORG` before enabling; however, **Iceberg v3 supports deletion vectors** ([S2]: "*Apache Iceberg v3 supports deletion vectors*")
- **Public Preview status** — not yet GA; breaking changes possible (low risk given timeline) ([S1])
- **Snowflake+Entra requires public networking** — cannot use Private Link for Entra OAuth ([S1]: "*must use public networking when authenticating with an Entra service principal*")

Investigation Findings

#	Finding	Result	Severity for Implementation

1	`external_access_enabled` is `false` on metastore	UC Admin must enable before any external Iceberg reads work	HIGH
2	No UniForm on any table — all at `minReaderVersion=1`, `minWriterVersion=2` (needs ≥2/≥7)	Protocol upgrade required per table; partially irreversible ([S2])	HIGH
3	No `EXTERNAL_USE_SCHEMA` grant — only SELECT, USE_SCHEMA, MODIFY, CREATE_TABLE	UC Admin must grant per-schema	HIGH
4	Metadata staleness — Iceberg metadata generated async, may lag Delta writes	Monitor `converted_delta_version`; `MSCK REPAIR TABLE` as fallback	MEDIUM
5	Network — TEST workspace reachable externally via PAT (validated 2026-03-24)	No Private Link needed for API access	RESOLVED
6	Protocol upgrade partially irreversible — reader/writer version + column mapping can't be undone ([S2])	Test on non-prod first; forward-compatible	LOW
7	Deletion vectors on existing tables — need REORG for Iceberg v2; v3 supports them natively ([S2])	Schedule during maintenance; REORG is idempotent	LOW

Investigation Status: COMPLETE

All investigation objectives have been met:

Objective	Status
Endpoint accessibility	CONFIRMED — 12 API endpoints tested, all respond
Network connectivity	CONFIRMED — reachable from external network
UC inventory captured	COMPLETE — 8 catalogs, ~66 schemas, ~466 tables, 3 SPs
UniForm readiness assessed	COMPLETE — zero tables ready, prerequisites documented
Permissions gap identified	COMPLETE — missing `EXTERNAL_USE_SCHEMA`
Cost analysis	COMPLETE — zero incremental licensing; real inventory costing above (< $1-30/mo depending on scenario)
"Why not read Delta directly?"	COMPLETE — 9-dimension comparison above
Implementation prerequisites	DOCUMENTED — 6 items for future ticket

References

All claims in this document were verified against official Microsoft Learn documentation, fetched 2026-03-23:

ID	Title	URL	Last Updated
[S1]	Access Azure Databricks tables from Apache Iceberg clients	https://learn.microsoft.com/en-us/azure/databricks/external-access/iceberg	2026-03-19
[S2]	Read Delta tables with Iceberg clients (UniForm)	https://learn.microsoft.com/en-us/azure/databricks/delta/uniform	2026-03-06
[S3]	Enable external data access on the metastore	https://learn.microsoft.com/en-us/azure/databricks/external-access/admin#external-data-access	—
[S4]	Databricks service principals / Auth	https://learn.microsoft.com/en-us/azure/databricks/dev-tools/auth/	—
[S5]	PyIceberg REST catalog configuration	https://py.iceberg.apache.org/configuration/#rest-catalog	—

[S6]	Apache Iceberg REST API spec	https://github.com/apache/iceberg/blob/master/open-api/rest-catalog-open-api.yaml	—
[K]	Project knowledge base	`../../knowledge.json` v1.14.0	2026-03-25
[DB]	Databricks TEST workspace API capture	`../../output/databricks/` (7 files, 987 KB)	2026-03-24